Predicting the Cost of Health Care

ISABEL ROMERO

Goals of the Investigation

- This study aims to investigate and determine the most influential factors on the cost of health care, which will make it possible to predict the overall cost.
- I used the Insurance dataset that was found on Kaggle.com.
- This investigation used a linear regression model in order to predict health care cost, as well as a decision tree.

What Factors Effect the Cost of Health Care

- All the data was found on Kaggle.com
- The data is about 1,300 lines long and studies many variables including:
  - Age
  - Sex
  - BMI
  - Amount of Children
  - Smoking
  - Region

Data Summary

```
      age            sex                 bmi           children
 Min.   :18.00   Length:1338        Min.   :15.96   Min.   :0.000
 1st Qu.:27.00   Class :character   1st Qu.:26.30   1st Qu.:0.000
 Median :39.00   Mode  :character   Median :30.40   Median :1.000
 Mean   :39.21                      Mean   :30.66   Mean   :1.095
 3rd Qu.:51.00                      3rd Qu.:34.69   3rd Qu.:2.000
 Max.   :64.00                      Max.   :53.13   Max.   :5.000
    smoker             region             charges
 Length:1338        Length:1338        Min.   : 1122
 Class :character   Class :character   1st Qu.: 4740
 Mode  :character   Mode  :character   Median : 9382
                                       Mean   :13270
                                       3rd Qu.:16640
                                       Max.   :63770
```

- The sex, region and smoker variables are dummy variables so they do not have a summary like the other variables.
- This summary shows the min, max, 1st quadrant, median, mean, and 3rd quadrant of the data.
- It shows a large gap between the minimum and maximum value of charges.

Methodology

- Multiple Linear Regression was used to analyze the impact that age, sex, BMI, number of children, smoking and region have on the overall cost of healthcare.
  - It also allowed me to remove factors that do not have a substantial impact in order to create a simpler model to predict the cost of healthcare.
- A decision tree allowed me to create a model to visualize the impact that each factor has on the overall charges. It shows how variations in the data lead to different healthcare charges.

Multiple Linear Regression

```
Call:
lm(formula = formula_0, data = Data_train)

Residuals:
     Min       1Q   Median       3Q      Max 
-11239.7  -2831.5   -986.7   1368.5  29916.4 

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)    
(Intercept)      -11752.11    1090.88 -10.773  < 2e-16 ***
age                 263.69      13.37  19.724  < 2e-16 ***
sexmale            -238.14     374.20  -0.636  0.52465    
bmi                 327.51      32.03  10.225  < 2e-16 ***
children            420.04     156.61   2.682  0.00743 ** 
smokeryes         23676.59     464.74  50.946  < 2e-16 ***
regionnorthwest    -562.41     537.34  -1.047  0.29550    
regionsoutheast    -819.28     544.07  -1.506  0.13241    
regionsouthwest    -967.22     535.04  -1.808  0.07093 .  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6078 on 1061 degrees of freedom
Multiple R-squared:  0.7486,	Adjusted R-squared:  0.7467 
F-statistic: 394.9 on 8 and 1061 DF,  p-value: < 2.2e-16
```

- This model shows that sex and region are not statistically significant variables.
- The variable with the largest intercept is smoking, which shows it is the most impactful on the cost of healthcare.
- The adjusted R-squared value of 0.7467 shows that the model is a good fit for the data.

New Regression Model

```
Call:
lm(formula = formula_1, data = Data_train)

Residuals:
     Min       1Q   Median       3Q      Max 
-11865.6  -2874.3   -922.9   1303.8  29577.8 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -12080.52    1040.18 -11.614   <2e-16 ***
age            263.99      13.33  19.804   <2e-16 ***
bmi            314.72      30.42  10.345   <2e-16 ***
children       414.77     156.34   2.653   0.0081 ** 
smokeryes    23661.15     461.66  51.253   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6078 on 1065 degrees of freedom
Multiple R-squared:  0.7476,	Adjusted R-squared:  0.7467 
F-statistic: 788.7 on 4 and 1065 DF,  p-value: < 2.2e-16
```

- The new regression model is a simpler model that does not contain the variables sex and region.
- It still shows smoking to be the strongest indicator of the cost of healthcare.
- It has an adjusted R-squared value of 0.7467, which shows this model is also a good fit for the data.

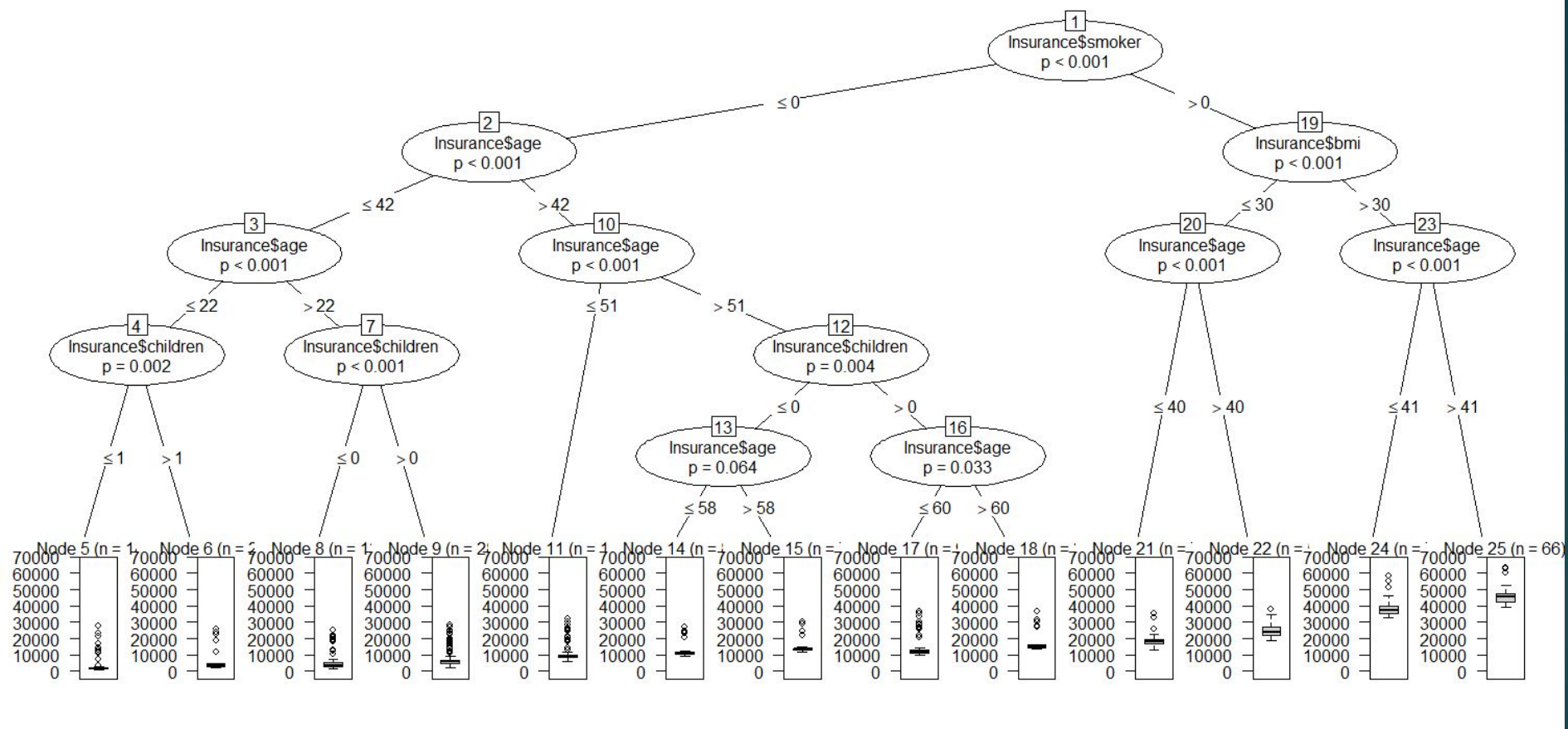
$$\hat{Y} = -12080.52 + 263.99(Age) + 314.72(BMI) + 414.77(Children) + 23661.15(Smoker)$$

Model Comparison

- The R-square value for the first model is equal to 0.7486.
- The RMSE value for the first model is 6018.55.
- The R-square value for the new model is equal to 0.7486.
- The RMSE value for the new model is 6035.24.

The R-squared values and RMSE values are very similar between the first model and the new model. Therefore, the new model is preferred because it is very accurate and simpler to use.

Decision Tree



- This shows the most important factor to be smoking.
- This decision tree is also limited to the confidence level 0.9 and minsplit equal to 100, which means a branch will only split if the sample size is at least 100.
  - These parameters make the decision tree simpler.

Conclusion

- The linear regression model and decision trees show that sex and region are not statistically significant indicators of health care cost, therefore they can be removed from the model.
- Both the linear regression model and decision tree shows that smoking has the largest impact on the cost of health care.
- The new model can be used to predict the cost of healthcare, with only age, bmi, children and smoking, which is a much simpler model.